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Master Thesis

"Brain Alignment as a Signal for Language Models: From Measurement to Application"

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Brain Alignment as a Signal for Language Models: From Measurement to Application

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Abstract

[Abstract pending condensation from the canonical rewrite.]

1 Introduction

[Pending.]

2 The Valid Training Claim

[Pending.]

3 Experimental Framework

[Pending.]

4 Results

[Pending.]

5 Discussion

[Pending.]

6 Limitations

[Pending.]

7 Conclusion

[Pending.]

Supplementary Material

A Evidence Records and Provenance

The E001–E030 identifier range covers the analyses and dispositions used in this thesis. Each existing record states what was executed, what its retained artifacts support, and which interpretation survived review. The main text states the resulting thesis claims. This appendix connects those claims to their final evidence owners, classifies every record identifier, identifies earlier readings that were superseded, and reports the retained path–hash pairs used for reproduction. Engineering-only, unused, unexecuted, and precondition-stopped records support only their stated disposition.

A.1 Evidence-record dispositions and claim ownership

Table 1 classifies every identifier once. The categories separate completed scientific records from restricted studies, stopped routes, design work, and unused identifiers.

Table 1: Final dispositions of evidence identifiers E001–E030.

Disposition	Records	Permitted use
Final executed scientific record	E008, E011, E016, E025, E030	Supports its settled scientific verdict.
Executed; interpretation corrected or narrowed	E002, E009, E026	Raw measurements remain fixed, while the learned-parameter, utility, or comparator-identification interpretation was narrowed on 2026-08-04.
Executed; interpretation corrected or narrowed	E003, E004, E005, E006, E010, E014, E015, E021	Supports only the corrected or scope-limited verdict.
Executed with a restricted diagnostic scope	E013, E017, E019, E020, E022	Supports only the stated local or diagnostic conclusion.
Precondition-stopped	E024	Supports the stopping decision; the intervention was not tested.
Deferred, analysis-only, or unauthorized	E012, E023, E027	Supports only its stated design, activation criterion, or non-authorization; no intervention result was produced.
Engineering or design record only	E001, E028, E029	Documents pipeline work or an outcome-blind design; supports no scientific result.
Unused identifier	E007, E018	Supports no scientific claim.

Table 2 follows the claim roles used in Sections **??** and **??**. Each row names one scientific role, the record that owns its final verdict, and the Results subsection where the evidence is interpreted.

Table 2: Scientific roles, final evidence owners, and settled conclusions.

Scientific role	Evidence owner(s)	Settled conclusion	Results owner
Controlled regional predictivity assay	E002	Controlled Tuckute region of interest (ROI) predictivity is supported under contiguous folds, nuisance subtraction, and an architecture-matched untrained control.	**??**
Controlled naturalistic voxelwise predictivity assay	E006	Trained representations outperform matched untrained representations for UTS03 and five held-out story blocks. The assay does not support claims about individual voxels or a population of participants.	**??**
Distillation-headroom analysis	E003; E015 supports the quality analysis	The cold GPT-2 route shows an observed quality-confounded gap; quality-aware warm-route headroom is unresolved and Qwen-route headroom is unmeasured.	**??**
Recorded-response intervention, participant-averaged estimand	E004 and E005; E010 and E014 bound the averaging interpretation	Fold-unit reanalyses of E004 and E005 supersede their earlier excludes-zero readings. E010 does not identify a participant-count response. E014 shows that averaging reduces response noise and emphasizes shared response while changing the estimand.	**??**
Recorded-response intervention, participant-specific estimand	E008; E011, E013, and E017 bound tested variants	No participant-general pairing advantage is demonstrated across the tested objectives and variants.	**??**
50-component linear target-projection measurability assay	E030	The fixed 50-component principal component analysis (PCA) target does not meet the internal $+0.002$ continuation rule. Aligned-minus-row-twin specificity remains unresolved.	**??**
Synthetic-target recovery assay	E016	WikiText assays support target uptake.	**??**
Retained-student movement audit	E025; E016 supports the earlier assay	Retained-student movement is supported at acceptable measured language model (LM) quality.	**??**
Comparator-adequacy audit and attribution assessment	E026; E016 supplies the original comparator	The saved text-derived auxiliary-control arm is non-identifying because pre-outcome target and KD-baseline comparability checks do not hold; post-training movement remains diagnostic.	**??**
Saved-student target-retention analysis	E030	The exact target is recoverable from the synthetic-response students on Tuckute, while incremental retention beyond ordinary knowledge distillation (KD) remains unresolved.	**??**
Composed-path diagnostic	E030	Within the declared linear pathway, predictive overlap, incremental overlap, and the aligned-over-row-twin increment are not demonstrated across participants. These diagnostics do not constrain the independent direct-transfer contrast.	**??**
Saved-student biological-transfer assay	E025; E026 bounds attribution	Participant-level biological transfer is not demonstrated. The direct synthetic-response-minus-KD contrast is near zero, while the text-derived comparison is non-identifying.	**??**

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Scientific role	Evidence owner(s)	Settled conclusion	Results owner
Practical-utility evaluation	E009	Mean out-of-domain ratios favor the recorded-response arm, but paired endpoint uncertainty was not reported; the prerequisite contrast is technically unstable, so brain-specific utility is not established.	**??**

A.2 Superseded interpretations

Some records were technically complete before their final interpretation was settled. Table 3 preserves the corrections that change how those records may be cited.

Table 3: Earlier interpretations and the verdicts that supersede them.

Records	Earlier interpretation	Final permitted reading
E003	KD uniquely destroys or preserves alignment.	The cold GPT-2 gap is quality-confounded; quality-aware warm-route headroom remains unresolved.
E004, E005, E008	Seed–fold cells provide independent uncertainty, and the averaged contrast excludes zero.	Fold-unit inference supersedes the cell bootstrap; E008 owns participant inference.
E006, E007	A voxel bootstrap supports population inference or a minimum detectable effect.	Those claims are withdrawn; the retained result concerns UTS03 and five held-out story blocks.
E010, E014	Nested participant subsets establish a monotone participant-count response.	Corrected random subsets do not identify a count response; E014 owns the response-reliability and estimand effects of averaging.
E015	A narrow-family best-layer correlation establishes a general scaling relation.	The full-range quality association survives; the capable-band relation and architecture residual remain uncertain.
E021	The apparent advantage is cognition-specific.	A phase-randomized shape twin matches the apparent advantage; the final finding is shape-specific.
E002	The 0.491 correlation ceiling can normalize unique R^2.	Correlation and variance-scale R^2 cannot be divided; the raw contrasts and verdict remain unchanged.
E013	The initial damaging variant establishes a stronger intervention result.	Only the anchored and contrastive tested branches inform the verdict; the larger expansion was not built.
E016, E025, E026	The defective permutation or the text-derived comparator matched only in dimension identifies brain-response content.	E025 owns the direct participant contrasts; E026 establishes that the auxiliary comparator is non-identifying.
E017	The aggressive forgetting configuration represents the full-tuning result.	The gentle matched-quality probe owns the scoped verdict.
E019	The local probe reproduces or refutes the external study.	It is a local lever diagnostic.
E020	The controlled residual closes the stimulus-predictable ceiling question.	Reference reliability fails, so the ceiling remains bounded but unresolved.
E030	Failure of the 50-component linear target projection explains the E025 result.	E030 limits the declared linear pathway without explaining or invalidating the independent direct-transfer result.

A.3 Retained load-bearing artifacts

Table 4 lists the verified subset of load-bearing path–hash pairs explicitly bound by an owning evidence record. It excludes smoke outputs, intermediate caches, prospective-design artifacts, and hashes not bound to exact filenames. The table reproduces the bindings for reader verification; the owning evidence record remains the authority for each artifact. Unbound E025 and E030 hashes and the E028 and E029 design artifacts are omitted. Appendices C, D, and F report the corresponding training settings, estimators and inference rules, and complete numerical and sensitivity tables.

Table 4: Retained paths and verified SHA-256 digests.

Record	Retained path and SHA-256
E002	outputs/E002_tuckute_feasibility.json SHA-256: ce45eb1786514a168e81248c426c472077a44245d8863ee5d0bbf6e4aacab44d
E003	outputs/E003_cold.json SHA-256: a30dceb63832a2fce838907cc9c782b4f6e8297c2a0136f94b5aacc9056edd6e
	outputs/E003_warm.json SHA-256: 6e75770387b0a84866060049d8fb8d3aecda898a95b2ce159ceffc226a5bf387
	outputs/E003_perplexity.json SHA-256: 2385e67ec61b7c64276f01e7b8e717e1d54cb1c1b23023715922e685cab5f97b
	outputs/E003_dissociation.json SHA-256: b7658aa0d1acf92bd62897aadec663aeda7a662abfa58890ca63eab6db5e5d38
E004	outputs/E004_brain_lever_Qwen.json SHA-256: 5f5582a651ecd86c25691fe24b6827d49a258c33e95127f630c522563d8dc6d4
E005	outputs/E005_Qwen.json SHA-256: e9b7be5b739370ed224de1bd8c6d4fa34d33881db308d3508e1edba66ddfcf6e
	outputs/E005_lambda_sweep_avg_Qwen_analysis.json SHA-256: 85013e6ff961aa88bd40e9e16dd19aaab3c89656f0dac8cd74c64560469f7938
E006	outputs/E006_lebel_gpt2.json SHA-256: ea424b7d2f8a32dd289e8ef8066c8609749f4edc1b4cdd132373a78bd6b6830b
	outputs/E006_lebel_Qwen.json SHA-256: 09b77b8801e937d8232ce7e7f16435e490f0b9300c78ce4ede23d00bf07d2581
E008	outputs/E008_g0.json SHA-256: fa2d18629e3818412a493539438d211c60cb8052de13286dd632983642d056cd
	outputs/E008_g1.json SHA-256: 0a60a95009f8bdb2c9aca1ba8f17f9e88f2f1be584a7db13bd35f6c9b51e6b5a
	outputs/E008_g2.json SHA-256: e51b58d2f4b19fa624ed7b46dd0e4bfe35d5fb5b57104d78aa06a269830aaecf
	outputs/E008_g3.json SHA-256: 70903174700a20b5eaf4b44f685a6e14317daff3f35c14b28d1d4ac645ac5803
E009	outputs/E009_a3_powered.json SHA-256: 1820c097a1a30d3a23bb0956221fbfb5b33c6dc67557e108ac6b79dc56c46afb
E010	outputs/E010_averaging_doseresponse.json SHA-256: 79312a649c3021df1fdc94aa46f3fcaf6c9b57fa30e59bd4fcd2e28db87ba5d0
	outputs/E010b_random_subsets.json SHA-256: 276d19bbf013f4f90d9ccbbd02151926a67fdc5af08dc5787902420d754f1d17
E011	outputs/E011_g0.json SHA-256: 299ca57c7733932d9272a44d06fe3cb6976ec535a887b5ad27072002426ea9e2
	outputs/E011_g1.json SHA-256: f3963b67cc01e0fae3807de354b613d4e6a3ca6ff8c6e55e1eed6b7c8b719bd3
	outputs/E011_g2.json SHA-256: 136449382dc627d5007df84b248b3c63d8b9bd4e2d1164c7399afbb9b867b4ce
	outputs/E011_g3.json SHA-256: 4da6639f3943c931beab6b7189fd6a03248161c846180a225e0e9c17af5b0745
E013	outputs/E013b_g0.json SHA-256: a98346f08ad41103d1cc7b9c28715dd810927c3fa09c63bf4d1db858ece19976
	outputs/E013b_g1.json SHA-256: e472ee523b6af62701472bca554f79eed64f319944549349b01d6b54d3b43346
	outputs/E013b_g2.json SHA-256: f9df4143b2b1cce4e871778f30a0b4f2bb48f561f4847693be0b5a4bd2c03361
	outputs/E013b_g3.json SHA-256: ae741c2c4e948356841e45b9053686823071365f8cfb9e362018b66213126cad
	outputs/E013v2_check.json SHA-256: a7ea8adbda9312c55192015f1b704948d13774fc2105937ac6b58d159c6e99de
	outputs/E013v2_lam3.json SHA-256: 35869107e166aade0097ef2e337d133a2ae98b27d7f494c1dc3a7b4b09c6a926
	outputs/E013v2_lam6.json SHA-256: dcb0ab4a31f21da71cf2aea87bdfdada8a1eb530d2955fa8923eccde57e3421a
E014	outputs/E014_averaging_encoding_seeded.json SHA-256: f16fbff6c9a86b65ac1ae6cc51a9ee9f9a1a82c1e3ef3aa4f0f17a7b980a5835
E015	outputs/E015_expand/E015_expand_merged.json SHA-256: 0aa4f149c1ae471f3b008ff77707570e2df9c4e3a5df7e61c189aa87f438abad
	outputs/E015_expand/panel_followup.json SHA-256: 6c5d9738e08028636a5bbbf21dfad1acde2da5756edeb3c628dd3f8664a6eefa

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Record	Retained path and SHA-256
E016	outputs/E016_tribe/phase3/phase3_combined_tribe_gpt2_n95999_s0-5_lam0.1.json SHA-256: 9fd5a657bde1d95cfad93a85a3e6513514c33b45d21d7245ef0a0336f7ee9d98
	outputs/E016_tribe/phase3/phase3_combined_textfeat_gpt2_n95999_s0-5_lam0.1.json SHA-256: 793af7beaab54aff1267e44ef16bdffa5b4f5854d8fc99bcfc99a2bbec0d382d
	outputs/E016_tribe/phase3/phase3_combined_tribe_vs_textfeat_s0-5_comparison.interpret_audit.json SHA-256: 2716234438f2b0a0519fab9d16bceebaffea2e1c6fc93e9cd6f90a6819df2f5e
	outputs/E016_tribe/phase3/phase3_rerun_tribe_gpt2_n95999_s0-2_lam0.1_save.tuckute_alignment.json SHA-256: f7ea1607f48bb6b91c9e9f4087ac691aca527ac4c46ddfecb758fd90554827f7
	outputs/E016_tribe/phase3/phase3_extra_tribe_gpt2_n95999_s3-5_lam0.1.tuckute_alignment.json SHA-256: cf5e7805bc5bd5fec4b6bd5df0d652b1b28368741c874d686c19ffa0c37096ac
	outputs/E016_tribe/phase3/phase3_combined_textfeat_gpt2_n95999_s0-5_lam0.1.tuckute_alignment.json SHA-256: fa94779f2b0a06c748ef9016b9494720d82c9b345ea9fb577bc4bde6ee898024
	outputs/E016_tribe/phase3/phase3_combined_tribe_vs_textfeat_s0-5_tuckute_alignment.interpret_recompute. json SHA-256: 9fe1a4253bc39b1b75ce5d25184332825706b95518a493532af3a276d10f10eb
E017	outputs/E017_matched_ppl_control/fullft_gentle_3subj.json SHA-256: 8f426e7b012f8461208bfb6a0e3e0bf3981eacf2feef91fb95f40638e5ca4e38
E019	outputs/E019_negi/results.json SHA-256: 983805956f37d8e17d81d3172113d46bdfe174c6ccce581ce5d1ae040d165758
	outputs/E019_negi/probe_strong.json SHA-256: 2fed0402fcce0067a81d81330289829c6fd1d6c5128dfb30cccda14f9787664b
	outputs/E019_negi/eval_posctrl.json SHA-256: c7a9a5f5c07b3802842c73dbde891f6e9185751f3898522dd13c8347c45c214f
E020	outputs/E020_eys/eys_story11_results.json SHA-256: d16d6f94f2f8f389db3178262bb399c93a8f973321684e1246bf1009f063acab
	outputs/E020_eys/eys_diagnose.json SHA-256: e8fab13e0fed3840a94fb9605d2ca4868ddc6e78d6cc975cede6f7f03b8579b0
	outputs/E020_eys/eys_diagnose2.json SHA-256: 84d268dc4a6211e7775cc5bbeca84e5d64eb52fbd8af963799011014eacb1879
E021	outputs/e021/arms_v5_results.json SHA-256: 95c05b8d259d82fb83c84d54c595176198a9ba9cfe50a70941c2cddd24d69b5b
	outputs/e021/arms_v5_shard_all.json SHA-256: b53bf1c88de36b397a9be753587e0e3846fde107c648eb496250850a1b7314e1
E022	outputs/e022/pilot_results.json SHA-256: a95a1e23be864322fce4f26ad215927b8fb74739ae35156c0da0091a0eccd24c
E024	outputs/e024/zuco_nr_reliability.json SHA-256: 34f412c1d40177510e2369296a223c00267da1aa3af940514ece47c3e5dfc4e2
	outputs/e024/positive_control.json SHA-256: acea6d09191cbd1e31062dd4c3ed1963de5ff95387f87f4a8d26b836edaa8e1a
	outputs/e024/gaze_richness_probe_NR.json SHA-256: b8eb0f32bd16e7eda77f957f23ad329c881ce2f550e9e5f4de0b48bef4ae606e
	outputs/e024/gaze_richness_probe_TSR.json SHA-256: 5d23870ad33f24a52044983446b347e51a9b6ee437c1e3659eaa633e8cc694a8
E025	outputs/E025/extraction.json SHA-256: b49c2795c55961cae161084afa1d3409dd662b893f9a40fa6a4e9373bc2c84c5
	outputs/E025/analysis.json SHA-256: 9a43410bfafced8130d286faa8c011c6e2097707979d476b5ce01d9ec44f106c
E026	outputs/E026/e026_audit.json SHA-256: dfac7498d89d488483f129ebd30f36013d98778a8304f0d3d97d9e0a8c1932ad
	outputs/E026/e026_independent_result_audit.json SHA-256: b2dca54c87efeca60c78a0de8287b4571991dcf3e653ce17dacc60583d2fda9f
E030	outputs/E030/tribe_tuckute_condition-b_n1000_full.npz SHA-256: a1fb8667e5c27a7e9bde35cb5f3a4f24622ad13c3b79e91a884c119ef3b65233
	outputs/E030/raw_scores.json SHA-256: cebea58cd7f7d82908b26a25714a1d256ab674a7b08d9a18d94835b0f7fcb6e4
	outputs/E030/analysis.json SHA-256: 43a4f3b01641072345149e651cdd0b1ae0c903bb74366db2006f31b89d8dd818

B Data, Responses, Targets, and Controls

This appendix specifies the data and target constructions behind the assays in Sections **??** and **??**. It moves from recorded-response inclusion and aggregation, through nuisance features and fold-local preprocessing, to synthetic and text-derived targets and their controls.

B.1 Data inclusion and response construction

The recorded-response analyses use two complementary datasets. The Tuckute resource provides isolated sentences and regional responses from multiple participants, whereas the LeBel resource provides continuous stories and voxelwise responses from one deeply sampled participant. Table 5 records the exact substrate and held-out partitions used by the two assay families.

Table 5: Included recorded-response material and held-out story folds.

Source or partition	Included material
Tuckute condition B	1,000 English sentences ordered by `item_id`; five left-hemisphere language ROIs: `AntTemp`, `IFG`, `IFGorb`, `MFG`, and `PostTemp`
LeBel fold 1	`adollshouse`, `adventuresinsayingyes`, `afatherscover`, `againstthewind`
LeBel fold 2	`alternateithicatom`, `avatar`, `backsideofthestorm`, `becomingindian`
LeBel fold 3	`beneaththemushroomcloud`, `birthofanation`, `bluehope`, `breakingupintheageofgoogle`
LeBel fold 4	`buck`, `catfishingstrangerstofindmyself`, `cautioneating`, `christmas1940`
LeBel fold 5	`cocoonoflove`, `comingofageondeathrow`, `exorcism`, `eyespy`

The Tuckute resource was acquired during silent sentence reading with 3-T functional magnetic resonance imaging (fMRI), a 2.0-s repetition time (TR), and 2-mm isotropic voxels [1]. This work inherits the released `response_target`; it does not reconstruct voxel time series or refit the functional localizers. In the released target, voxel responses were z-scored within scanning session and then averaged within each ROI. The remaining operations select condition B, order rows by `item_id`, and construct either an averaged or participant-specific response matrix.

The participant-averaged construction combines five original training participants cell by cell. Their unique identifiers (UIDs) are 848, 853, 865, 875, and 876. A missing contributor is skipped, but the resulting sentence-by-ROI matrix must be complete before analysis. This response is used by the controlled regional assay and the participant-averaged recorded-response intervention. It estimates predictivity for the shared response of this fixed group because averaging occurs before readout fitting and before held-out R^2 is calculated.

Participant-specific construction retains one complete sentence-by-ROI matrix per person. UID 853 is excluded because 60 entries are missing and the five-ROI grid is incomplete. The inclusion rules differ because cell-wise averaging can use an available entry, whereas participant-level inference requires a complete sentence-by-ROI matrix for each person. UID853 has 60 missing sentence–ROI cells: those 60 participant-averaged cells use four contributors, while the other 4,940 use all five. This construction distinction does not assume that the missing entries are ignorable for every estimand. The remaining participants are UIDs 797, 837, 841, 848, 856, 865, 875, 876, and 880. Within this set, {848, 865, 875, 876} form the inherited training-participant subgroup and {797, 837, 841, 856, 880} form the held-out-participant subgroup. Appendix D specifies the fold and seed aggregation; the nine retained participants are the biological inference units. Thus participant averaging changes both the signal-to-noise ratio and the estimand; it does not create additional independent participants.

The naturalistic assay uses participant UTS03 from the LeBel story-listening resource [2]. The responses were acquired at a nominal TR of 2.0 s. The present implementation places the stimulus features and released response matrices on the 2.0045-s grid used by the dataset adapter. Each released HDF5 `data` matrix contains one row per scanner sample and 95,556 cortical-voxel columns. This work begins from those matrices and does not rerun reconstruction, motion correction, or cortical registration.

The 20 assay stories follow their recorded order and are divided into the five contiguous four-story folds in Table 5. Complete stories are held out, which prevents temporally dependent samples from the same story from crossing the evaluation boundary. The separate story `wheretheressmoke` is excluded from model fitting and scoring and is used only for voxel reliability selection.

Reliability is estimated from the ten released repeats of `wheretheressmoke`. With random seed 0, each of 50 random splits assigns five repeats to each half. Responses are averaged within each half, correlated voxelwise across the 291 scanner samples, and corrected with the Spearman–Brown formula. The mean corrected correlation across the 50 splits is the voxel's reliability estimate. Voxels with reliability strictly greater than 0.5 are retained, leaving 11,442 voxels. Because the repeated story is independent of the 20 assay stories, this screen does not select voxels from an evaluation outcome. The resulting estimand remains held-out-story prediction within UTS03; the retained voxels are spatial measurements, not independent biological replicates.

B.2 Stimulus, nuisance, and fold-local preprocessing

Each assay separates the contextual model feature from lower-level or static stimulus features that could explain recorded responses without the contextual representation under study. Table 6 summarizes these blocks before the fold-specific transformations are applied.

Table 6: Contextual and nuisance feature blocks by assay.

Assay	Contextual block	Low-level nuisance block	Static or additional block
Tuckute	Mean-pooled activations of the declared LM layer over non-padding token positions	Sentence token length and normalized item position	Mean noncontextual input embedding; released imageability in the exact-substrate primary design
LeBel	Final-subtoken hidden state for each word under its available left context	Word rate, phoneme rate, mean word duration, word length, and log word frequency	985-dimensional `english1000` lexical-semantic features

In the controlled regional assay, the historical E002 implementation recomputed the static embedding block for each trained or untrained LM. The trained score therefore remains controlled relative to its implemented nuisance set, but the trained-minus-untrained gap does not isolate learned parameters against a byte-identical static nuisance block. The saved-student biological-transfer assay instead fixes the static block to GPT-2 Medium so that every student arm is evaluated against the same nuisance representation. The 50-component linear target-projection measurability assay adds the released imageability covariate in its primary design; Appendix F reports the design without imageability as a continuity sensitivity. GPT-2 XL surprisal and probabilistic context-free-grammar surprisal provide a secondary stress test because both contain model-derived contextual information that overlaps the construct being measured.

The Tuckute analyses use five contiguous outer folds of 200 sentences. For each fold, stimulus-domain scaling and contextual and static PCA are fitted on the other 800 sentences and applied unchanged to the held-out block. Contextual and static feature blocks are each reduced to rank 50, equalizing their nominal dimensions before readout fitting, while the scalar nuisances remain unprojected. The synthetic brain-response target has one fixed exception: its coordinate standardization is learned once from the external WikiText training corpus, as described below. Any target PCA used in a Tuckute assay is still fitted only on the corresponding 800 outer-training rows.

The LeBel preprocessing begins with the time-aligned word and phone annotations in the TextGrid files. Word duration is the successive word-onset difference clipped to 0–3 s, with 0.3 s assigned to the terminal word. Contextual word features are extracted in overlapping 256-token windows with stride 128; each word receives the hidden state of its final subtoken under the maximum available left context. Word length, log word frequency, word duration, and `english1000` features are Lanczos-resampled with window 3 to the 2.0045-s grid. Word and phoneme rates are direct counts in TR-centered histogram windows.

After these stream-specific operations, the first 10 and final 5 samples of each story are removed with the fixed slice `[10:-5]`. The contextual, low-level, and static feature streams then receive within-story z-scoring and finite-impulse-response copies at delays one through four TRs. The recorded blood-oxygen-level-dependent (BOLD) responses are z-scored voxelwise within story but receive no finite-impulse-response expansion. For a held-out story, this transformation uses that story's own response mean and variance before scoring. This is a shared transductive normalization across model conditions, so the estimand is prediction of within-story standardized responses rather than deployment to a story whose response moments are unavailable. Feature and response rows are then truncated to their common aligned length. Within each outer fold, the contextual and static blocks receive train-story PCA to rank 100, while the low-dimensional nuisances remain unprojected. A predictor scaler fitted on the training stories then z-scores the assembled design. The same folds, nuisance arrays, temporal transforms, and response rows are used for trained and architecture-matched untrained LMs. Appendix D specifies response centering, ridge selection, fresh-readout fitting, aggregation, and statistical inference.

B.3 Synthetic brain responses and text-derived auxiliary targets

Target construction. The synthetic-response route uses a fixed WikiText-103 corpus that is disjoint from the Tuckute sentences. The corpus builder reads the training split of `Salesforce/wikitext`, configuration `wikitext-103-raw-v1`, in dataset order. It removes blank lines and document headers, splits each remaining line after sentence-final punctuation, retains sentences of 4–45 whitespace-delimited words, and removes case-insensitive duplicates and exact case-insensitive matches to Tuckute sentences. The first retained sentences form the fixed training file and the following sentences form the untouched held-out file. The E003 corpus build recorded 96,000 training sentences and 2,000 held-out sentences before the later target-cache filtering step. The E016 target caches contain 95,999 training sentences and 1,999 held-out sentences after that step; these counts describe two retained builds rather than one drifting split.

TRIBE v2 maps each sentence to synthetic brain responses on a standardized cortical surface with 20,484 fsaverage5 locations [3]. The target builder loads `facebook/tribev2`, sets `data.text_feature.model_name` to `unsloth/Llama-3.2-3B`, and sets `data.features_to_use` to `[text]` in the runtime configuration. The Llama-3.2-3B component is TRIBE's internal text encoder; it is distinct from the frozen GPT-2 Medium distillation teacher. The published TRIBE description does not list the Tuckute sentence set among its training datasets. This supports dataset separation at the level of reported provenance, but it is not a categorical proof that no overlapping text or derivative data entered any upstream component.[3] For this thesis, the sentence-level target is the average of TRIBE's one-second output samples for that sentence. This operation creates one training vector per sentence; it is not a claim of physical alignment to the acquisition timing of the Tuckute fMRI responses. Its `synthetic-text` configuration assigns each token a 0.35-s event followed by a 0.05-s gap, extends timelines shorter than 1.0 s, and uses a 1.0-s TR. Each sentence has a separate timeline, and its retained one-second predictions are averaged coordinatewise to form one 20,484-dimensional vector. The same construction is applied to Tuckute item IDs 1–1,000, with `item_id=item_index+1`.

Let Y_{ij} be synthetic response coordinate j for training sentence i. Each coordinate is standardized using the training-corpus mean $\mu_{j,\mathrm{train}}$ and standard deviation $\sigma_{j,\mathrm{train}}$:

$$\widetilde{Y}_{ij} = \frac{Y_{ij} - \mu_{j,\mathrm{train}}}{\sigma_{j,\mathrm{train}} + 10^{-6}}.$$

This operation centers and scales each cortical coordinate using only the synthetic-response training corpus. The same stored moments are applied to held-out sentences, and the small constant prevents division by a near-zero scale. The moments and standardized targets are computed and stored in float32. The stored training target retains all 20,484 standardized coordinates. Appendix C specifies how the training objective uses them; no target PCA is applied before that use. When a later assay requires a lower-dimensional target, its PCA basis is learned only from the relevant outer-training rows.

The text-derived auxiliary target preserves the target width and standardization procedure while replacing the TRIBE construction with frozen LM features. Let $h_i \in \mathbb{R}^{1024}$ be the mean-pooled layer-12 representation of sentence i extracted from the same frozen GPT-2 Medium model used as the distillation teacher, with at most 64 token positions. Let $R \in \mathbb{R}^{1024\times 20484}$ be drawn once with `numpy.random.default_rng(0)`, with entries $R_{ab} \sim \mathcal{N}(0,1)$, and then frozen. The target for sentence i is

$$T_i = \frac{h_i R}{\sqrt{1024}}.$$

The projection gives the auxiliary target the same 20,484-coordinate width as the synthetic brain-response target. The factor $1/\sqrt{1024}$ keeps its scale from growing mechanically with the 1,024 input dimensions. Each projected column is standardized by the same formula, using its own mean and standard deviation computed on the WikiText training corpus.

Pairing controls. The controls differ in which relationship they disrupt. The following constructions make the preserved and disrupted relationships explicit.

The recorded-response intervention constructs its permutation only from the current 800-sentence training complement. The participant-averaged route uses one draw per fold and seed, whereas the participant-specific route fits five independently permuted controls per fold and seed and averages their scores. The control randomly reorders ten contiguous response blocks within the 800-sentence training complement. It disrupts sentence–response pairing while preserving the response-value multiset and within-block order, but it is not constrained to move every block.

The saved synthetic block permutation and the frozen row twin serve different purposes. Because the legacy block permutation was not a derangement and truncated unequal block pairs, it can retain fixed rows and duplicate or omit rows; it is therefore only a sensitivity. The retained manifest does not record an exact fixed-row count, so the thesis does not quantify or treat this control as an identifying comparator. A derangement is a permutation with no fixed points, so every stimulus is paired with another stimulus's target. The frozen row-twin control instead uses Sattolo's algorithm to construct one derangement inside each 200-item outer block before target values or participant scores are read. Its deterministic seed is derived from the block identifier. For each outer fold, aligned TRIBE rows define one train-only 50-component PCA basis, and the same basis is applied to the twin. This comparison isolates row identity at matched target-row content and linear target capacity. It concerns one frozen row-twin control and therefore does not establish invariance across possible derangements or identify a causal brain mechanism.

Table 7 summarizes the construction and attribution role of each target or control.

Table 7: Synthetic targets and pairing controls used in the thesis.

Construction	Matched or preserved property	Attribution role
Synthetic brain-response target	Sentence order, TRIBE geometry, standardization, and width	Synthetic-response supervision under study
Text-derived auxiliary target	Sentence order, width, standardization, and corpus	Frozen GPT-2 Medium projection controlling for auxiliary supervision derived from text
Recorded-response permutation	Response values, marginal distribution, and within-block order	Randomly reordered sentence–response pairing; fixed blocks remain possible
Saved block-permutation sensitivity	Synthetic target values and approximate block structure	Legacy pairing sensitivity; fixed, duplicated, or omitted rows remain possible
Frozen row-twin control	Exact within-block row multiset and train-only 50-component PCA capacity	Row identity under one deterministic derangement with no fixed points

The text-derived auxiliary target matches the synthetic target on row order, output width, standardization procedure, and corpus. The two targets do not share a source representation, and matching width and standardization does not establish matched covariance, effective rank, recoverability, or optimization difficulty. Those differences limit the attribution question the comparison can answer; Section **??** and Appendix F report the measured audit. With the data, targets, and controls fixed, Appendix C specifies how the training objectives use them.

C Training Interventions and Reproduction

This appendix specifies the optimized objectives, the parameters reached by each gradient, and the student state available after training. Appendix B owns the construction of the data, targets, and controls; Appendix D owns fresh readouts and statistical inference; and Appendix A owns verified artifact paths and hashes. Section **??** and Appendices E–F own the resulting observations and verdicts. Temporary target heads never enter a later evaluation. A student checkpoint is retained only when the run explicitly saves one; otherwise evaluation uses the frozen in-memory student and only the result artifact survives.

C.1 Ordinary KD and language-quality rules

Let $z_T(x)_t$ and $z_S(x)_t$ be teacher and student logits at a non-padding causal-language-model position t. At temperature τ, their softened token distributions are

$$p_T^\tau(\cdot \mid x, t) = \mathrm{softmax}(z_T(x)_t/\tau)\,, \qquad q_S^\tau(\cdot \mid x, t) = \mathrm{softmax}(z_S(x)_t/\tau)\,.$$

The implemented output-distillation loss is

$$\mathcal{L}_{\mathrm{KD}} = \tau^2 \frac{\sum_{x,t} m_{xt} D_{\mathrm{KL}}(p_T^\tau(\cdot \mid x, t) \,\|\, q_S^\tau(\cdot \mid x, t))}{\sum_{x,t} m_{xt}}, \tag{1}$$

where m_{xt} is one at non-padding positions and zero otherwise. Equation 1 averages teacher-to-student token-level Kullback–Leibler (KL) divergence over valid positions. The factor τ^2 compensates for the gradient-scale change introduced by temperature softening. Every executed output-distillation route uses $\tau = 2$, gradient-norm clipping at 1, and AdamW.

The language objective is shared within a paired intervention, but its initialization and quality role differ across families. Table 8 records those differences.

Table 8: Language objectives and quality rules by training family.

Family	Teacher, student, and initialization	Language objective and corpus	Quality rule
Distillation-headroom analysis	GPT-2 Medium teacher; GPT-2 Small student initialized either from pretrained weights or randomly	Output KD on 96,000 WikiText sentences; 2,000 fixed held-out sentences	Held-out perplexity is a convergence measure; the warm and random-start routes differ in initialization and training duration
Recorded-response intervention (sentence)	Qwen2.5-0.5B student; Qwen2.5-1.5B teacher when the route is teacher-anchored	Masked next-token cross-entropy in the unanchored screen; output KD in anchored routes; fixed WikiText quality probe	The recorded-response arm and permuted-response control share the scoring implementation; ordinary KD comparisons require comparable perplexity
Recorded-response intervention (naturalistic)	Qwen2.5-0.5B student; frozen Qwen2.5-1.5B teacher in anchored routes	Unanchored or output-KD-anchored language path, according to the executed grid	The full-parameter route uses its declared forgetting criterion on the language probe
Synthetic-response intervention	GPT-2 Medium teacher; pretrained GPT-2 Small student shared within seed	Output KD on 95,999 WikiText sentences; 1,999 fixed held-out sentences	Each target arm must satisfy Equation **??**; saved-student comparisons also apply the within-seed bits-per-byte rule

The headroom family uses seeds 0–2, batch size 16, and maximum length 64. The pretrained student is trained for one epoch at 5×10^{-5}, whereas the random-start student is trained for two epochs at 3×10^{-4}. These are separate optimization regimes, not an initialization-only contrast. The recorded- and synthetic-response settings are specified with their target paths below.

C.2 Recorded-response intervention

For a recorded response y_i, the common sentence-level target path is

$$x_i \longrightarrow h_{12}(x_i) \longrightarrow W h_{12}(x_i) \longrightarrow \mathcal{L}_{\mathrm{target}}.$$

Here h_j denotes entry j in the model's returned hidden-state sequence, whose entry 0 is the embedding output. The thesis shorthand "hidden-state index 12" therefore names the exact attachment returned by the implementation. The vector $h_{12}(x_i)$ is mean-pooled over non-padding token positions, and W is a temporary linear target map unless the selected loss is readout-free.

The common joint objective is

$$\mathcal{L} = \lambda_{\mathrm{lang}}\mathcal{L}_{\mathrm{lang}} + \lambda_{\mathrm{target}}\mathcal{L}_{\mathrm{target}}, \qquad \lambda_{\mathrm{lang}} = 1. \tag{2}$$

The first term preserves the declared language objective, while the second exposes the retained student path to the recorded response. For teacher-anchored routes, $\mathcal{L}_{\mathrm{lang}} = \mathcal{L}_{\mathrm{KD}}$; the unanchored screen instead uses masked next-token cross-entropy.

For predictions $\widehat{Y}$ and targets Y with minibatch size B and target width D, the squared-error target loss is

$$\mathrm{MSE}(\widehat{Y}, Y) = \frac{1}{BD}\sum_{b=1}^{B}\sum_{j=1}^{D}\left(\widehat{Y}_{bj} - Y_{bj}\right)^2.$$

This loss averages equally over minibatch rows and target coordinates. Consequently, every reported target weight is interpreted under this normalization.

The common sentence-level regime places rank-16 low-rank adaptation (LoRA) adapters with scaling 32 and dropout 0.05 in the model-appropriate attention and feed-forward projections [4]. The pretrained base, input embeddings, and tied output matrix are frozen; the adapters and a co-trained W are optimizer parameters. The target gradient reaches W and the adapters that produce h_{12}, while adapters after the attachment state receive only the language-loss gradient. After training, the adapters are merged into the student and the temporary target map is discarded. The merged student is frozen before a fresh evaluation readout is fitted as specified in Appendix D.

Table 9 states only the implementation change relative to this common path.

Table 9: Recorded-response intervention variants.

Variant	Change from the common target path	Executed grid	Retained object
Co-trained mean squared error (MSE)	A new W is trained jointly with the producing adapters; correctly paired and permuted responses use the same path	Rank 16; seeds 0–2; 3 epochs; 2×10^{-4}; batch 16; target weight 10. Participant-averaged training uses one permutation draw; participant-specific training uses five	Merged student; W discarded
Frozen readout	W_0 is ridge-fitted on untuned outer-training features with $\alpha = 100$ and remains fixed during intervention training	Same sentence screen grid and paired-permutation construction as its matched comparison	Merged student; W_0 discarded
Contrastive	Symmetric InfoNCE compares normalized Wh_{12} with the paired five-ROI response using in-batch negatives [5]	Rank 16; seeds 0–1; 3 epochs; target weight 1; temperature 0.07; three permutation draws	Merged student; W discarded
Increased capacity	The co-trained MSE path is unchanged; adapter rank increases from 16 to 64	Seeds 0–1; 6 epochs; target weight 10; five permutation draws; remaining settings inherited from the participant-specific grid	Merged student; W discarded
Naturalistic voxelwise	A participant-specific head maps pooled 25-word segments to reliable voxels after a 4-s response delay	UTS01–03; 14 tuning and 6 evaluation stories; unanchored and output-KD-anchored grids recorded in the owning configurations	Merged student; voxel head discarded

Continued on next page

Table 9 (continued)

Variant	Change from the common target path	Executed grid	Retained object
Full parameter	The voxelwise MSE and output-KD anchor update all student parameters except the tied input/output embedding; no LoRA is used	UTS01–03; seeds 0–2; 2 epochs; 10^{-5}; batch 16; target weight 10	Student used for immediate frozen evaluation; voxel head discarded; no reusable student checkpoint retained

The exploratory sentence screen changes only $\mathcal{L}_{\text{target}}$. Cosine and squared-Pearson losses use a co-trained target map. Linear centered kernel alignment (CKA) is readout-free: it compares the minibatch geometry of h_{12} and the response matrix directly, so no temporary W participates in that branch [6]. These are implementation variants of the recorded-response intervention; their outcomes remain in Section **??** and Appendix E.

C.3 Synthetic-response intervention

The synthetic-response intervention uses the same joint objective in Equation 2, but the attachment and trainable path differ. For GPT-2 input x, the retained student path is

$$x \longrightarrow e_{\text{tok}}(x) + e_{\text{pos}}(x) \longrightarrow h_6(x) \longrightarrow h_{7:12}(x) \longrightarrow q_S^{\tau}(\cdot \mid x),$$

and the temporary target branch is

$$\text{meanpool}(h_6(x)) \longrightarrow W_{\text{tmp}}\, \text{meanpool}(h_6(x)) \longrightarrow \mathcal{L}_{\text{target}}.$$

As above, h_6 names hidden-state index 6 in the returned sequence, not an informal count of visible blocks. The output-KD gradient reaches the entire student path. The target gradient reaches W_{tmp}, token and positional embeddings, normalization components, and student blocks that produce h_6; blocks after the attachment receive only the output-KD gradient. Because GPT-2 ties its token embedding and output matrix, that retained matrix receives the target gradient through its input role and the output-KD gradient through both roles.

The teacher is frozen, whereas every student parameter and W_{tmp} is optimized. All target arms map the 768-dimensional mean-pooled attachment state to all 20,484 coordinates using the same mean MSE. Appendix B owns the construction and standardization of the synthetic brain-response target, the text-derived auxiliary target, and the saved block permutation. The common optimization grid uses 95,999 training sentences, one epoch or 24,000 optimizer steps, learning rate 5×10^{-5}, batch size 4, and maximum length 64.

Table 10: Synthetic-response training arms and retained states.

Arm	Additional supervision	Shared optimization settings	Saved state
Ordinary KD arm	None; $\lambda_{\text{target}} = 0$ and no temporary target map	Pretrained GPT-2 Small initialization, frozen GPT-2 Medium teacher, corpus, output-KD loss, and seeds 0–5	Student checkpoint only when artifact saving is enabled; otherwise result artifact
Synthetic-response arm	Correctly paired synthetic brain-response target through W_{tmp}; $\lambda_{\text{target}} = 0.1$	Same initialization, teacher, corpus, schedule, and seeds 0–5 as the matched ordinary KD arm	Temporary map always discarded; student saved only under the configured artifact path
Saved block-permutation sensitivity	Saved block-permuted synthetic target through the same W_{tmp} and mean MSE	Same initialization, teacher, corpus, schedule, target weight, and seeds 0–5	Temporary map discarded; result artifact retained; student saving follows the run configuration
Text-derived auxiliary-control arm	Text-derived auxiliary target through the same target-map shape and mean MSE; $\lambda_{\text{target}} = 0.1$	Same pretrained student, teacher, corpus, and schedule; executed for seeds 0–5	Temporary map discarded; seed-specific students saved when requested

The text-derived auxiliary-control arm was first executed for seeds 0–2 and later extended prospectively to seeds 3–5. The combined six-seed analyses use seeds 0–5; earlier three-seed tables retain only the initial run for provenance.

At the final optimizer step, the temporary target map is always discarded. When no checkpoint is requested, synthetic-target recovery uses the frozen in-memory student and only the resulting measurements are stored. When artifact saving is enabled, later assays reload the retained causal-language-model checkpoint and still fit a fresh readout; the temporary training map is never reused. Thus training can shape the retained student through the target gradient, but the training head itself cannot compute a later target-recovery or biological-transfer score.

The frozen row-twin control is not a training arm. It is constructed only for the later 50-component linear target-projection measurability assay described in Appendix B. Comparator adequacy and all arm outcomes are reported in Section **??** and Appendix F.

C.4 Replay scope and provenance handoff

Replay requires four linked objects: the locked software environment, the executed entry point and arguments, the declared seed and final-state rule, and the retained result or student artifact. Table 11 provides the compact index. It is not a command-complete substitute for the owning result configurations or the verified path–hash ledger in Appendix A.

Table 11: Training entry points and retained replay information.

Family	Executed entry point	Final-state and retained-output rule	Provenance handoff
Distillation-headroom analysis	`scripts/run_kd_alignment.py`	Final warm or random-start student evaluated in memory; result bundle records configuration and quality measurements	Verified retained bundles are indexed in Appendix A
Recorded-response intervention (sentence)	`scripts/run_brain_lever.py`; target losses in `scripts/brain_loss.py`	Final merged student evaluated after each fold–seed cell; temporary target map discarded; no intermediate quality frontier retained	Primary and participant result shards are indexed in Appendix A
Recorded-response intervention (naturalistic)	`scripts/run_lebel_tune.py`	Final merged or full-parameter student evaluated immediately; temporary voxel head discarded	Recorded grids and result bundles remain under their owning configurations; verified paths are delegated to Appendix A
Synthetic-response intervention	`scripts/run_tribe_phase3.py`	Final student evaluated in memory; optional checkpoint written only when a save directory is supplied; temporary target map discarded	Target, saved-student, and analysis bundles are indexed in Appendix A

The current replay environment uses the Python requirement in `pyproject.toml` and the exact package and CUDA build resolved by `uv.lock`. These files are current replay pins; they are not complete historical environment manifests for every earlier run. Result JSONs retain model identifiers and executed training configurations, and later saved-student bundles bind selected weights, inputs, runners, and analyses by hash. Many early runs do not record immutable upstream Hugging Face revision identifiers, so exact historical replay additionally depends on a retained model cache or a verified saved-student artifact. This limitation must be preserved rather than filled with an inferred revision.

Smoke runs, logs, temporary caches, throughput probes, and model-loading checks are diagnostic outputs and do not receive manuscript-level path–hash rows. Appendix A owns the verified artifact ledger and any stated omissions. With the optimization and replay scope fixed, Appendix D specifies how the retained students are evaluated and how uncertainty is assigned.

D Estimators, Aggregation, and Inferential Scope

Section **??** states the design-level estimands and inference units. This appendix answers four narrower questions: how the predictive quantities are related, how fitting and aggregation are nested, how uncertainty and decision rules are assigned, and where information-theoretic analogies stop. Appendix B owns the inputs and preprocessing definitions, Appendix C owns the training objectives and retained model states, and Section **??** owns the empirical outcomes. The quantities below are assay-conditional predictive contrasts. They are neither causal effects nor direct estimates of information-theoretic objects.

D.1 Predictive estimands and ridge estimation

For response coordinate j, Section **??** defines the held-out score $R_j^2(M)$ for a readout design M in Equation **??**. Let N denote the declared nuisance design and $F = [N, X]$ the full design after adding contextual LM representations X. The nuisance-only and full scores are $R_j^2(N)$ and $R_j^2(F)$, and the reported semipartial increment is

$$\Delta R^2_{\mathrm{semi},j}(X) = R^2_j(F) - R^2_j(N).$$

This difference asks how much held-out predictive value is gained by adding X to the nuisance design. Because the nuisance-only and full ridge readouts are separately regularized and evaluated out of sample, the difference can be negative. It is not a nonnegative variance decomposition.

Population partial R^2 answers a different question. Let $\mathcal{R}^2_{N,j}$ and $\mathcal{R}^2_{F,j}$ be the population coefficients of determination for nested, unregularized linear projections on N and F. When $\mathcal{R}^2_{N,j} < 1$,

$$R^2_{\mathrm{partial},j} = \frac{\mathcal{R}^2_{F,j} - \mathcal{R}^2_{N,j}}{1 - \mathcal{R}^2_{N,j}} = 1 - \frac{\mathrm{Var}(Y_j - \Pi_F Y_j)}{\mathrm{Var}(Y_j - \Pi_N Y_j)}, \tag{3}$$

where Π_N and Π_F are the corresponding population linear projections. Equation 3 measures the fraction of variance left after the nuisance projection that is removed by adding X. The semipartial quantity instead measures the gain relative to total outcome variance. The thesis reports the cross-validated semipartial increment, not a transformed cross-validated partial coefficient.

Aggregation across target or response coordinates defines another distinction. For J held-out coordinates, let SSE_j and SST_j denote the coordinate-specific residual and total sums of squares. The two aggregations used in the thesis are

$$\overline{R^2}_{\mathrm{equal}} = \frac{1}{J}\sum_{j=1}^{J}\left(1 - \frac{\mathrm{SSE}_j}{\mathrm{SST}_j}\right), \qquad R^2_{\mathrm{pooled}} = 1 - \frac{\sum_{j=1}^{J}\mathrm{SSE}_j}{\sum_{j=1}^{J}\mathrm{SST}_j}. \tag{4}$$

The first expression gives every coordinate equal weight. The second weights coordinates by their held-out variance, so a nearly constant coordinate cannot contribute as much as a variable coordinate. The WikiText synthetic-target recovery assay uses equal-coordinate averaging, whereas the saved-student target-retention analysis pools sums of squares across the fixed target coordinates. Neither calculation treats coordinates as independent repetitions.

Ridge regression supplies a regularized linear predictor. Let $y \in \mathbb{R}^n$ be a centered response vector, let $M \in \mathbb{R}^{n\times p}$ be the centered and standardized predictor matrix, and let $\beta \in \mathbb{R}^p$ be its coefficient vector. If

$$y = M\beta + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, \sigma^2 I), \qquad \beta \sim \mathcal{N}(0, \tau_\beta^2 I),$$

then the posterior mode minimizes

$$\|y - M\beta\|_2^2 + \lambda\|\beta\|_2^2, \qquad \lambda = \frac{\sigma^2}{\tau_\beta^2}.$$

This is a maximum a posteriori interpretation only under the stated Gaussian likelihood, isotropic zero-mean prior, and scaling convention. The fitted linear predictor represents the conditional mean only when that mean is linear and the model is correctly specified.

The sentence-level Tuckute and WikiText readouts use the candidate penalties

$$\{1, 10, 100, 1000, 10000\}.$$

Each multi-output fit selects one scalar penalty on its training partition by ridge cross-validation; it does not select a separate penalty for every response or target coordinate. The naturalistic LeBel readouts use

$$\{10, 10^2, 10^3, 10^4, 10^5, 10^6, 10^7\},$$

again selecting one multi-output penalty from training stories only.

The E026 comparator audit uses

$$\{0.1, 1, 10, 100, 1000\}$$

and selects one penalty by aggregate held-out R^2 across both target families and both fixed coordinate samples within five contiguous inner folds. These selection rules are nested inside the corresponding training partition; the untouched outer block or held-out corpus is used once for evaluation. In the executed assays, λ is selected from a finite grid using the relevant training complement; operationally, it is predictive regularization rather than a fixed scientific prior. Neither this derivation nor a positive held-out score makes the fitted coefficients causal, identifies a biological mechanism, or shows that an LM and the brain implement the same computation.

D.2 Cross-validation, aggregation, and inference units

The governing leakage rule is simple: every data-adaptive operation that can use stimulus-domain information is fitted on the relevant training complement. Fixed protocol operations may be applied before the split, but learned centering, scaling, dimensionality reduction, penalty selection, and readout fitting remain inside it. Appendix B specifies the exact fixed preprocessing and split definitions.

The naturalistic assays require one explicit distinction. Feature resampling, edge slicing, within-story feature and BOLD z-scoring, and finite-impulse-response delay copies are fixed story-local operations. Because the held-out story's response normalization uses that story's own moments, the resulting estimand is conditional on this normalization protocol. Across-story PCA, assembled-design scaling, response centering, ridge selection, and readout fitting are learned only from the readout-training stories. The 50-component linear target-projection measurability assay has one other declared exception: it reuses target-standardization moments fixed during target construction, while its target PCA and readouts remain fold-local.

Table 12 records the held-out boundary, learned operations, and final inference unit for each scientific role.

Table 12: Cross-validation nesting and inference units by scientific role.

Scientific role	Held-out boundary	Operations learned inside the boundary	Aggregation and final unit
Controlled regional predictivity assay	Five contiguous Tuckute folds, each with an 800-sentence training complement and a 200-sentence test block	Centering, scaling, PCA, one multi-output ridge penalty, and the response readout	Declared ROIs and folds are averaged; the sentence set is fixed, and untrained-model seeds are technical repetitions
Distillation-headroom analysis	The same five Tuckute folds for brain predictivity and a fixed WikiText probe for language quality	Brain-response transforms and readouts are fitted inside each fold; quality scores use the fixed scoring protocol	Model condition is the descriptive unit, seeds index stochastic arms, and model families are the resampling clusters for the cross-family quality association
Recorded-response intervention	The student trains on each 800-sentence outer complement; the untouched 200-sentence block is divided into five inner readout folds	Inner-fold transforms, penalty selection, and the fresh response readout	Participant-averaged analyses end at five shared outer folds; participant-specific analyses aggregate seed–fold cells within each participant
Controlled naturalistic voxelwise predictivity assay	Five complete-story folds; voxel reliability uses a separate repeated story	Training-story PCA, assembled-design scaling, response centering, one multi-output ridge penalty, and the readout	Story-block signs and voxel summaries describe within-participant robustness; they do not create participant inference
Naturalistic recorded-response intervention	Complete stories remain held out from fresh evaluation readouts	Training-story transforms, ridge selection, and participant-specific voxel readouts	Seeds are technical repetitions within three descriptive participant probes
Synthetic-target recovery assay	The student trains on 95,999 WikiText sentences and is evaluated on 1,999 fixed held-out sentences	The fresh target-readout penalty is selected using training data	Equal-coordinate target R^2 is compared within six matched technical seeds

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Table 12 (continued)

Scientific role	Held-out boundary	Operations learned inside the boundary	Aggregation and final unit
Saved-student biological-transfer assay	Frozen students enter five contiguous Tuckute 800/200 response-readout folds	All representation transforms, nuisance transforms, penalties, and fresh response readouts use the 800-sentence complement	Six seeds and five folds are aggregated within each of nine participants; participants are the biological units
50-component linear target-projection measurability assay	Five contiguous Tuckute folds; aligned targets and the frozen row twin share the train-fold target-PCA basis	Each aligned or twin multi-output readout selects one scalar penalty on its own training complement	Folds are aggregated within each of nine participants; there is no student-seed axis
Saved-student target-retention analysis	Frozen students and targets use the same five outer folds	Representation and target transforms and the multi-output target readout remain train-fold-local	Sums of squares are pooled across target coordinates; paired contrasts end at six technical seeds
Composed-path diagnostic	Both linear paths are fitted on the same outer-training rows and scored once on the untouched test block	Target-to-response and student-to-target coefficient blocks are fitted separately inside each fold	Folds and six seeds are aggregated within each of nine participants
Practical-utility evaluation	Fixed in-domain and out-of-domain corpora are scored without fitting a biological readout	No evaluation-time model fitting; corpus-level losses are computed under the fixed scoring protocol	Paired arm contrasts end at eight technical seeds and remain conditional on the fixed models and corpora

The aggregation order can be written without treating technical repetitions as new scientific units. Let $d_{u,r}$ be a matched contrast for final unit u and technical cell r, and let $\mathcal{A}_u$ denote the declared within-unit aggregation:

$$\widehat{\theta}_u = \mathcal{A}_u(\{d_{u,r} : r \in \mathcal{R}_u\}), \qquad \widehat{\Theta} = \mathcal{I}\Big(\widehat{\theta}_1, \ldots, \widehat{\theta}_U\Big). \tag{5}$$

Here $\mathcal{A}_u$ may be a mean, a median, or pooled sums of squares, while $\mathcal{I}$ is the descriptive summary or inferential procedure applied to the U final units. Equation 5 says that seeds and folds are first reduced within a participant when the participant is the scientific unit. Analyzing every seed–fold cell as if it were an independent participant would change both the estimator and its uncertainty.

The final unit determines what can generalize. Seeds vary optimizer randomness under one fixed procedure. Stimulus folds share observations and have overlapping training complements. Voxels and ROIs share a participant, preprocessing, and stimuli. Target coordinates share one generated target and student. None can replace a participant when the claim concerns people [7]. Conversely, several participants evaluated on one fixed sentence set do not by themselves establish generalization to a population of stimuli. Shared-fold and story-block summaries are therefore robustness checks, not substitutes for participant or stimulus-population inference.

D.3 Uncertainty, multiplicity, and decision rules

Four choices must remain separate. The inference unit identifies the observations that enter uncertainty. The interval or test describes variation across those units. The multiplicity family identifies which simultaneous claims require adjustment. The decision rule states what evidence was required to continue to a later stage. Unless stated otherwise, the reported intervals are two-sided 95% intervals.

For U final-unit estimates $\widehat{\theta}_1, \ldots, \widehat{\theta}_U$, the ordinary t interval is

$$\overline{\theta} \ \pm \ t_{1-\alpha/2, U-1} \frac{s_\theta}{\sqrt{U}}, \qquad \overline{\theta} = \frac{1}{U} \sum_{u=1}^{U} \widehat{\theta}_u. \tag{6}$$

Here α is the two-sided error rate. Equation 6 uses the number of final units U, not the number of coordinates, folds, or technical runs. Its sampling interpretation requires independent final-unit estimates and approximate normality when U is small. Participant intervals are conditional on the fixed stimuli, trained students, and analysis protocol used to construct each participant estimate.

For K matched technical contrasts $d_1, \ldots, d_K$, the exact two-sided magnitude-preserving sign-flip test used in the seed-level analyses is

$$p_{\text{flip}} = \frac{1}{2^K} \sum_{\epsilon \in \{-1,+1\}^K} \mathbf{1}\left[\left| \frac{1}{K} \sum_{k=1}^{K} \epsilon_k d_k \right| \geq \left| \frac{1}{K} \sum_{k=1}^{K} d_k \right| \right]. \tag{7}$$

This test requires the contrast signs to be exchangeable under the null conditional on their magnitudes. It differs from the participant sign test, which uses only the number of positive and negative participant effects and requires independent participant signs. The Wilcoxon signed-rank sensitivity additionally assumes independent and approximately symmetric paired effects. The shared-fold cluster bootstrap resamples the five outer-fold summaries as clusters. Because the folds reuse observations through overlapping training complements, both that bootstrap and the fold t interval are fixed-block stability summaries rather than stimulus-population intervals. The cross-family quality bootstrap instead resamples model families rather than individual models.

Table 13 maps each scientific role to its uncertainty procedure and stage-specific decision scope.

Table 13: Uncertainty procedures and decision rules by scientific role.

Scientific role	Interval or test and its unit	Multiplicity and decision scope
Controlled regional predictivity assay	Held-out-fold dispersion and trained–untrained contrasts on the fixed sentence set	Prespecified layer and nuisance grids bound the assay; folds do not support participant or stimulus-population inference
Controlled naturalistic voxelwise predictivity assay	Within-participant trained–untrained point contrasts and five story-block signs; no voxel-resampling interval	Spatial voxels are not independent units, and overlapping story folds remain robustness checks
Distillation-headroom analysis	Fold–seed bootstrap conditional on convergence; cross-family quality association uses a family-cluster bootstrap, with pooled Fisher intervals only as a sensitivity	The declared retention bands classify opportunity within this assay and are not universal alignment thresholds
Participant-averaged recorded-response intervention	Seeds are averaged within each outer fold before a five-fold t interval	The interval describes stability across the five shared blocks of the fixed sentence set
Participant-specific recorded-response intervention	Participant medians over seed–fold cells receive a participant t interval and exact one-sided sign test; a shared-fold interval and cluster bootstrap assess block sensitivity	Requiring agreement between participant and shared-fold readings is a conservative conjunctive rule, not a joint crossed-population confidence interval
Naturalistic recorded-response intervention	Seed summaries are technical repetitions within three participant-specific probes	The small participant set supports descriptive mechanism probing, not population inference
Synthetic-target recovery assay and retained-student movement audit	Six matched-seed contrasts receive descriptive intervals and declared exact sign-flip tests	Manipulation and language-quality criteria qualify later interpretation but provide no biological replication
Comparator-adequacy audit and attribution assessment	Fixed-sample margins and paired arm contrasts are reported without coordinate-level p-values	Comparator adequacy is a pre-outcome or baseline design condition for attribution; realized movement is diagnostic, and statistical precision cannot repair a non-identifying comparator

Continued on next page

Table 13 (continued)

Scientific role	Interval or test and its unit	Multiplicity and decision scope
Saved-student biological-transfer assay	Nine participant effects receive a t interval, a one-sided upper bound, an exact sign test, and a Wilcoxon sensitivity	Seeds and blocks remain descriptive; the upper bound is compared only with the prospectively inherited continuation rule
50-component linear target-projection measurability assay	Participant effects receive t intervals and exact sign tests	Its participant quantities belong to a separate Bonferroni family with a simultaneous one-sided bound
Saved-student target-retention analysis	Six paired technical-seed contrasts receive a descriptive interval and an exact 2^6 sign-flip test	Target retention has its own adjusted interval family and does not create biological evidence
Composed-path diagnostic	Participant path-product quantities receive t intervals and exact sign tests	The composed path has its own adjusted interval family; it is predictive and is not a causal mediation effect
Practical-utility evaluation	Eight paired technical-seed endpoint observations were recorded, but the completed analysis reports only mean contrasts and an observed-variance sensitivity calculation	No paired endpoint interval or equivalence test was retained; the minimum detectable effect (MDE) supplies no hypothesis or equivalence decision

For the primary retained-student movement audit, let H_s and H_s^{KD} be layer-6 representation matrices for the same 1,999 untouched WikiText sentences in technical seed s, and let $C = I - \mathbf{1}\mathbf{1}^\top/n$ center examples. The centered relative-Frobenius distance and linear-CKA distance are

$$D_{F,s} = \frac{\|C(H_s - H_s^{\mathrm{KD}})\|_F}{\|CH_s^{\mathrm{KD}}\|_F}, \qquad D_{\mathrm{CKA},s} = 1 - \frac{\|(CH_s)^\top(CH_s^{\mathrm{KD}})\|_F^2}{\|(CH_s)^\top(CH_s)\|_F\,\|(CH_s^{\mathrm{KD}})^\top(CH_s^{\mathrm{KD}})\|_F}. \tag{8}$$

The first quantity measures centered activation change relative to the ordinary-KD activation norm. The second measures the change in centered linear representation geometry, with zero denoting identical linear-CKA geometry. Global parameter displacement is reported separately as

$$D_{\theta,s} = \frac{\|\theta_s - \theta_s^{\mathrm{KD}}\|_2}{\|\theta_s^{\mathrm{KD}}\|_2}. \tag{9}$$

These post-training quantities diagnose the realized intervention; they are not comparator-identification conditions.

An MDE describes the resolution of a planned contrast under its declared variance model, sample size, error rate, and power. It does not show that a smaller effect is zero or that a larger effect matters biologically. Likewise, language-quality criteria decide whether two arms are sufficiently comparable for a specified analysis; they do not establish a biological endpoint.

The $+0.002$ unique-R^2 rule was inherited prospectively to decide whether participant-level biological transfer justified a matched-control follow-up, then frozen as a reference for the 50-component linear target-projection measurability assay. It can support a statement about whether an assay-specific simultaneous bound reaches that continuation rule. It cannot define biological importance across another response scale, nuisance model, layer, cohort, or target. No retrospective choice of a more favorable analysis can change that scope.

D.4 Information-theoretic boundaries

Population partial R^2 has an information-theoretic identity only under restrictive assumptions. For scalar Y, jointly Gaussian (X, Y, Z), correctly specified population-linear conditional means, and unregularized population residual variances,

$$I(X; Y \mid Z) = -\frac{1}{2}\log\bigl(1 - R^2_{\mathrm{partial}}\bigr), \qquad \Delta\mathcal{R}^2_{\mathrm{semi}} = \bigl(1 - \mathcal{R}^2_N\bigr)\,R^2_{\mathrm{partial}}. \tag{10}$$

Equation 10 converts a population residual-variance ratio into conditional mutual information under the stated Gaussian model. The multivariate analogue uses a log-determinant ratio. The thesis instead estimates finite-sample, cross-validated semipartial differences from separately regularized ridge fits. Those differences may be negative and depend on the nuisance design, rank, penalty, and held-out substrate, so they are neither estimates nor lower bounds of $I(X;Y \mid Z)$ [8].

A data-processing claim requires a separate conditional-independence premise. Let S be stimulus text, B a recorded brain response, $T = g(S)$ the deterministic synthetic brain-response target, and H the retained student state. The chain

$$B \longrightarrow T \longrightarrow H \qquad \Longleftrightarrow \qquad B \perp H \mid T$$

would require all information from B that reaches H to pass through T, which would imply $I(B;H) \leq I(B;T)$. The executed design does not establish this chain because S directly supplies both T and H, and it also drives B. Conditioning on S changes the question: because T is deterministic given S, $I(T;B \mid S) = 0$. That identity does not show that the target is useless. It shows that conditioning on the exact stimulus cannot establish additional target information beyond that stimulus or recover the desired unconditioned mediation claim. Data processing therefore supplies a bound only after the relevant conditional independence is defended; it does not predict the sign of a held-out encoding contrast [8].

Output matching and auxiliary-target fitting impose still weaker constraints on retained representations. Let d_H be any declared discrepancy between teacher and student internal states. Then, without additional identifying assumptions,

$$D_{\mathrm{KL}}(p_T^{\tau} \| q_S^{\tau}) = 0 \;\not\Rightarrow\; d_H(H_S, H_T) = 0, \qquad \mathcal{L}_{\mathrm{target}} \downarrow \not\Rightarrow\; \Delta R^2_{\mathrm{transfer}} > 0.$$

The first non-implication holds because different internal parameterizations can realize the same output distribution. The second holds because a temporary target map can absorb part of the auxiliary objective, and a predictable target can be learned without producing participant-general improvement on recorded responses. Low output or target loss can establish optimization success under its own objective. It cannot replace fresh tests of retained-student movement, comparator adequacy, brain-response attribution, biological transfer, or practical utility. Section **??** reports those empirical stages without treating any formal analogy as their outcome.

E Supporting Analyses and Stopped Routes

This appendix separates secondary evidence from work that did not reach an interpretable scientific test. The distinction matters because a working pipeline, a restricted local diagnostic, and a stopped route answer different questions. Only executed analyses can bear on the thesis claims, and even then only within the scope of the design that produced them. Table 14 states the status language used below. Appendix A provides the corresponding provenance records, while Appendix F reports the full numerical results and sensitivity analyses. The headline empirical verdicts remain in Section **??**.

Table 14: Evidence-status categories used in this appendix.

Category	Reader question	Permitted contribution	Excluded claim
Pipeline validation	Did the pipeline behave correctly when the expected signal was known?	Supports implementation and analysis checks.	No claim about recorded brain responses or training benefit.
Supporting analysis	Does an alternative specification change a settled interpretation?	Tests a named objection within its recorded design.	No broader claim beyond the tested specification.
Restricted local diagnostic	Does a reduced local implementation recover a reported effect?	Reports whether the effect appears under the available local conditions.	No reproduction, refutation, or cross-study equivalence claim.
Failed instrument	Could the proposed measurement support the intended inference?	Identifies why the instrument was not interpretable.	No biological null from a measurement failure.
Precondition-stopped route	Did the design satisfy the prerequisite required before intervention?	Records why stopping prevented an uninterpretable downstream attribution claim.	No result for the intervention that was not run.
Analysis-only or prospective design	Was a future test screened or specified without opening its outcome?	Clarifies the intended identifying question and activation criterion.	No empirical evidence for the proposed mechanism.

E.1 Pipeline validation and scientific scope

The end-to-end pipeline was first tested in a setting where the expected signal was known. Real Pereira sentences [9] were paired with synthetic responses generated from teacher features, named nuisance variables, and noise. The test exercised ordinary and target-guided KD, the temporary target head, middle-layer extraction, the contiguous held-out split, fold-local ridge fitting, nuisance subtraction, and aggregation across optimization seeds. The expected teacher–student ordering appeared, the nuisance component remained small, and excessive target weight degraded the result rather than receiving an automatic advantage.

These observations rule out several basic implementation failures. They show that the held-out evaluation tail was not used for target-guided training, that the scoring pipeline could recover a planted feature–response relation, and that the analysis did not reward every target-guided run by construction. They do not establish that an LM predicts recorded brain responses, that the nuisance set is sufficient for real stimuli, or that a brain-response objective improves a student. The synthetic test therefore validates the end-to-end implementation, not the scientific claim.

E.2 Response averaging and recorded-response robustness

Two averaging analyses addressed different questions. The first changed how many participants contributed to the intervention target. Because each target was rebuilt from a different participant subset, increasing the subset size also changed participant identity, response reliability, and target construction. After correcting the subset analysis, the apparent improvement did not form a stable monotonic dose–response curve. The result therefore does not identify a causal benefit of adding more participants to the training target.

The second analysis held the frozen representation and encoding procedure fixed while averaging increasing numbers of recorded responses. Controlled brain predictivity increased as averaging suppressed idiosyncratic and measurement noise and emphasized the shared stimulus-evoked component. This is a valid signal-to-noise gain for a group-response estimand. It changes what is being predicted, however, and does not supply participant replication or rescue the individual-response intervention.

The remaining analyses tested whether specific design choices concealed a recorded-response benefit. Table 15 maps each change to its objection and scope.

Table 15: Recorded-response robustness analyses and their scope.

Objection	Design change	Supported disposition	Remaining open
Auxiliary weight was poorly chosen.	Repeated the recorded-response intervention over the tested weight range.	No stable target-specific gain appeared within that range.	Other objectives or schedules could behave differently.
Adapter capacity was too limited.	Increased the tested low-rank adaptation capacity.	The tested capacity increase did not recover the effect.	This does not test every stronger manipulation.
Pointwise regression was the wrong loss.	Replaced it with a contrastive response objective.	The alternative loss did not produce a reliable target-specific benefit.	Other response objectives remain possible.
Regional targets hid a voxelwise effect.	Tested the available naturalistic voxelwise training procedure.	The procedure did not establish the required manipulation.	The broader multi-participant full-parameter route was not built.
Parameter-efficient tuning was too restrictive.	Tested full-parameter tuning, including a gentler three-participant probe.	Aggressive tuning caused forgetting; the gentler probe showed no reliable target-specific gain.	Three participants do not support population inference.

These analyses weaken the listed specific alternative explanations. They do not establish that every recorded-response objective must fail. Section **??** owns the participant-level intervention verdict, and Appendix F reports the numerical variants.

E.3 Reduced external and cross-modal probes

External protocols were used as local diagnostics only when their decisive ingredients could be approximated with the available data and models. The standard is therefore narrower than reproduction: the question is whether a reported effect appears under the local conditions, not whether a reduced mismatch can confirm or refute the source study.

The first diagnostic adapted the differentiable temporal-head and target-specificity logic of Negi et al. [10] to the available LeBel responses [2] and decoder-only LMs. Under the gentle schedule, the retained representations changed little; stronger schedules caused severe language degradation. The local runs therefore did not establish a target-specific manipulation over their controls. The comparison nevertheless differs from the source study in population, response substrate, model family, language, training scale, and control construction. It is therefore a restricted local failure to establish the target-specific manipulation, not a reproduction or refutation of the published result.

The second diagnostic adapted a speech-based protocol from Moussa et al. [11]. Starting from the same Wav2Vec2-base model family, the design required a positive phoneme-prediction gain before constructing a matched-quality specificity test. In the reduced single-participant setup, target loss decreased but the phoneme-gain criterion was not met, so the downstream specificity comparison was not built. The source study used a larger, multi-participant setting with different timing and model choices. The result is thus a reduced local non-reproduction of the prerequisite effect, not evidence against the published claim.

Neither diagnostic supplied an external positive control for the present intervention pipeline. They narrow what the local pipeline recovered while leaving the external literature open under its original conditions.

E.4 Failed instruments and precondition-stopped routes

Three routes stopped because an upstream measurement or attribution condition failed. In each case, continuing to intervention would have produced a number without a defensible interpretation.

The shared-response route asked whether the predictable group response could serve as a ceiling for participant-specific intervention effects. In principle, this reference estimates the expected response shared across participants for the same stimulus, $m(S) = \mathbb{E}[Y \mid S]$. In practice, the leave-one-participant reference was not reliable enough, while temporally adjacent samples preserved apparent residual predictivity through leakage and autocorrelation. Once a clean temporal gap was imposed, the controlled residual was bounded but could not close the ceiling question. The ceiling instrument therefore failed; the analysis does not show that participant-specific structure is absent or that a universal biological ceiling has been reached.

The reading-time route asked whether a cognitive target could provide a simpler intervention signal than fMRI. Raw and residual reading-time targets initially produced similar frozen-probe learning advantages over simple controls. A phase-randomized control preserved the target's marginal distribution and temporal structure, however, and matched those advantages. The design therefore did not isolate cognition-specific content from target shape. This attribution failure does not imply that reading time contains no useful signal; it means that the available contrast could not identify that signal.

The gaze route required three conditions before privileged-information training: a measurable target, evidence that it added information beyond text, and enough independent groups for leakage-resistant evaluation. Natural-reading gaze was measurable but not informative under the tested assay. Task-directed gaze was more predictable, but the signal could reflect task intent or label leakage, and the number of independent paragraph groups was too small for a reliable group split. The intervention therefore stopped at its preconditions. No corresponding claim was tested for electroencephalography (EEG).

Stopping at these failed prerequisites prevents later results from being interpreted as identified intervention effects. They identify why the proposed instruments could not answer their questions, but they do not turn measurement failure into a biological null.

E.5 Prospective designs without empirical evidence

The remaining routes specify questions that would be useful only after their activation conditions are met. They have no weight in the empirical verdicts of this thesis.

The same-lineage alignment–quality design would compare models from one lineage along an alignment–quality frontier, then test paired ordinary fine-tuning where alignment changes while language quality remains approximately stable. An analysis-only slope-and-knee screen did not identify a high-quality flat regime that justified the intervention. No paired training test was authorized, so the screen contributes no evidence for or against objective-specific shedding.

The matched auxiliary-control design would ask whether any intervention benefit came from brain-response content rather than generic target geometry, learnability, optimization pressure, or retained-student movement. The planned control would match those properties while removing brain-response content. Its activation criterion was not met, and the saved text-derived auxiliary control was not valid for that role. No target or student from this branch supports a scientific comparison.

The cross-modal falsification design would compare real training fMRI, a phase-randomized response control, and a geometry-matched control on an independent biological endpoint. The route remains on design hold because the exact external protocol depends on unavailable author artifacts; proceeding would require explicit authorization of a clearly labeled reimplementation. Only outcome-blind metadata and pipeline development were completed: no endpoint data were inspected, no student was trained, and no biological score was computed. The planned analysis was intended to predict the sign and ordering of those outcomes before inspection. It remained contingent on an informative cross-modal regime; only its algebraic consistency check was completed, without independent review or execution.

These designs clarify what stronger tests would need to identify. They remain prospective and do not alter the thesis conclusions.

F Claim-Relevant Sensitivities and Verification Tables

This appendix reports the verification detail needed to judge whether the main conclusions depend on an analysis choice or an unstable aggregate. Section **??** owns the decisive observations and scientific verdicts; Appendix A owns the visible evidence-record crosswalk and the verified subset of path–hash bindings. Every value below traces to its owning evidence record, including values whose artifacts are not listed in that verified subset.

The tables preserve the distinction between inference units. Participant-level effects support inference across the sampled biological units under the assumptions in Appendix D. Held-out blocks diagnose dependence on the fixed stimulus partition, and optimization seeds diagnose the stability of a fixed training procedure. Neither blocks nor seeds create additional biological or model-population replicates.

F.1 Claim-relevant analysis choices

An analysis choice can affect a thesis claim in four ways. An *interpretation-changing* analysis replaces an earlier estimand or inference unit. A *scope-bounding* sensitivity narrows what the primary result supports without reversing its frozen decision rule. A *confirmatory* check leaves the scoped interpretation unchanged. A *limitation or correction* removes authority from a design element rather than generating a more favorable result. Tables 16 and 17 apply these labels to the analyses that matter for the thesis claims.

Table 16: Interpretation-changing and scope-bounding analyses.

Analysis choice	Scientific role	Status	Effect on interpretation
Representation rank	Distillation-headroom analysis	Scope-bounding	The ordering of the model arms survived, but absolute magnitudes and some signs changed. Headroom is conditional on the declared rank.
Auxiliary weight and language quality	Participant-averaged recorded-response intervention	Scope-bounding	Near-quality-matched settings remained uncertain across folds. The clearest fold-positive setting incurred a substantial language-quality cost.
Quality band and model family	Distillation-headroom analysis	Scope-bounding	The full-range association was -0.783; within the capable-model band it was -0.501, with interval $[-0.859, +0.188]$. Language quality remains a required control, not a deterministic explanation.
Response construction and inference unit	Participant-specific recorded-response intervention	Interpretation-changing	Participant-specific targets and participant-first inference answer a different question from an averaged target and fold-level summary. The change removed support for a participant-general gain.
Participant and subgroup deletion	Saved-student biological-transfer assay and 50-component linear target-projection measurability assay	Scope-bounding	Removing the predeclared highest-baseline participant reversed the relative transfer point estimate. Every leave-one-participant-out linear target-projection mean remained negative.
Held-out stimulus block	Controlled naturalistic voxelwise predictivity assay; recorded-response intervention; saved-student biological-transfer assay; 50-component linear target-projection measurability assay	Scope-bounding	The naturalistic assay remains a one-participant result. No leave-one-block-out intervention analysis produced a positive participant-mean contrast that changed the participant-level disposition, and the linear target-projection result survived every block deletion.
Controlled-predictivity contrast by seed	Practical-utility evaluation	Scope-bounding	Only $5/8$ technical-seed signs were positive. The utility analysis therefore follows a seed-unstable prerequisite rather than a reliably induced target-specific change.

Table 17: Confirmatory checks and design limitations.

Analysis choice	Scientific role	Status	Effect on interpretation
Layer, nuisance set, and target aggregation	Saved-student biological-transfer assay and 50-component linear target-projection measurability assay	Confirmatory	Prespecified nearby choices did not overturn the primary rules. The 50-component linear target projection did not meet its continuation rule; retention and composed-path quantities remain diagnostics of that pathway.
Saved-student optimization seed	Synthetic-target recovery assay; retained-student movement audit; saved-student target-retention analysis; saved-student biological-transfer assay; composed-path diagnostic	Confirmatory	Seed agreement can establish reproducible optimization or manipulation, but it cannot substitute for participant agreement.
Saved controls audited after training	Comparator-adequacy audit and attribution assessment	Limitation or correction	The saved block-permutation sensitivity is defective, and the text-derived auxiliary control fails pre-outcome and KD-baseline comparability. Neither can identify an effect of brain-response content.

Table 18: Corrected classification of the frozen comparator audit.

Role	Total	Pass or within	Fail or outside	Interpretation
Identification checks	37	20	16	One additional check is unresolved; failures include KD-baseline recovery and headroom, covariance geometry, effective rank, and nuisance predictability.
Post-training diagnostics	18	4	14	Parameter, Frobenius, and CKA margins describe realized intervention differences and do not determine comparator validity.

The original combined engineering count, $30/55$, is retained only as provenance for the frozen audit.

The interpretation-changing analysis altered the question being answered; it was not a more favorable reanalysis of the earlier averaged-target result. The remaining sensitivities either narrowed the scope, confirmed stability near a frozen specification, or exposed a limitation. None licenses replacing a prospective primary estimator or decision rule with a more favorable variant.

F.2 Participant-level intervention contrasts

The participant-specific recorded-response estimator first forms a paired target-specific contrast for participant u, held-out block k, and optimization seed s, where $P = 5$ is the number of matched permuted-target draws.

$$d_{uks} = R^{2,\text{unique}}_{\text{recorded},uks} - \frac{1}{P}\sum_{p=1}^{P} R^{2,\text{unique}}_{\text{perm}(p),uks}, \qquad e_u = \text{median}_{k,s}\, d_{uks}, \qquad \widehat{\theta}_{\text{part}} = \frac{1}{U}\sum_{u=1}^{U} e_u. \tag{11}$$

Equation 11 subtracts the mean matched permuted-target score inside every seed–block cell, then aggregates the crossed cells within participant before averaging participants. The resulting U =9 participant effects, each based on 15 paired cells, are the biological inputs to the cohort estimate.

Table 19: Participant-specific recorded-response intervention contrasts.

UID	Participant effect e_u
797	$+0.00002885$
837	$+0.00100412$
841	-0.00018259
848	-0.00063374
856	-0.00037809
865	-0.00058994
875	$+0.00070575$
876	$+0.00011067$
880	$+0.00083826$
Mean	$+0.00010$

The signs are mixed around a small cohort mean. The table supports participant-first aggregation and heterogeneity inspection; it does not imply that every participant has exactly zero effect.

The saved-student biological-transfer assay uses a different target and within-participant aggregator. Let T_u, X_u, and K_u denote the participant-level scores for the synthetic-response target, text-derived auxiliary control, and ordinary KD. The three displayed contrasts satisfy

$$\Delta_u^{T-X} = T_u - X_u, \qquad \Delta_u^{T-K} = T_u - K_u, \qquad \Delta_u^{X-K} = X_u - K_u, \qquad \Delta_u^{T-X} = \Delta_u^{T-K} - \Delta_u^{X-K}. \tag{12}$$

Equation 12 shows why a positive relative contrast against the auxiliary control need not imply improvement over ordinary KD: the auxiliary control may instead have moved downward. Seeds and outer folds are averaged within each participant before these contrasts are formed.

Table 20: Participant-level saved-student biological-transfer contrasts.

UID	Δ_u^{T-X}	Δ_u^{T-K}	Δ_u^{X-K}
797	-0.00023801	-0.00005311	$+0.00018490$
837	$+0.00227168$	$+0.00226591$	-0.00000577
841	$+0.00000452$	-0.00078813	-0.00079265
848	-0.00002346	-0.00002667	-0.00000320
856	-0.00022167	-0.00036255	-0.00014087
865	-0.00064994	-0.00108762	-0.00043767
875	$+0.00009116$	-0.00006339	-0.00015455
876	-0.00013023	-0.00014342	-0.00001319
880	$+0.00012305$	$+0.00013517$	$+0.00001212$
Mean	$+0.000136$	-0.000014	-0.000150

The transfer contrasts are also heterogeneous. The relative mean Δ^{T-X} is small and positive, while the direct mean Δ^{T-K} is -0.000014; removing the predeclared highest-baseline participant changes the relative point estimate to -0.000131. These rows explain why the relative comparator cannot replace the direct contrast or participant-first inference.

The two participant tables must not be pooled or compared row by row. They use different targets, contrasts, and within-participant aggregators; the shared identifiers only show that they were evaluated in the same participant cohort.

F.3 Participant-level mechanism diagnostics

The target-projection diagnostic evaluates the declared 50-component linear pathway on the same sentence substrate; it is not a prerequisite for the independent direct-transfer contrast. For each participant, A_u measures the target's unique response predictivity above nuisance, and Q_u compares that score with the frozen row-twin control. These quantities compare target-above-nuisance predictivity with sentence-pairing specificity before the later target-retention analysis and composed-path diagnostic are interpreted.

Table 21: Participant-level linear target-projection diagnostics.

UID	Target above nuisance A_u	Aligned minus frozen twin Q_u
797	-0.00266759	-0.00031777
837	-0.00448823	-0.00785204
841	-0.02042083	$+0.00588154$
848	-0.00001541	$+0.00158126$
856	-0.01394534	-0.00066703
865	$+0.00208088$	$+0.00684375$
875	$+0.00266981$	$+0.01519279$
876	-0.00660269	-0.00590127
880	$+0.00293994$	$+0.00106256$
Mean	-0.004494	$+0.001758$

The mean target-above-nuisance score is -0.004494, and every leave-one-participant-out mean remains negative. This first prerequisite misses its frozen practical rule. The aligned-minus-frozen-twin values have substantial effects of both signs, so that paired specificity question remains unresolved rather than contradicting the first result.

The later diagnostics ask whether the part of the target that a student predicts overlaps with the target direction that predicts participant responses. For participant u, O_u is the absolute composed student-to-target-to-response overlap, D_u is its brain-target-minus-KD increment, and J_u is the aligned-minus-frozen-twin increment. All three average optimization seeds and outer folds within participant before participant-level inspection.

Table 22: Participant-level composed-path quantities.

UID	Absolute overlap O_u	Synthetic response minus KD D_u	Aligned minus twin J_u
797	$+0.00045651$	$+0.00018799$	$+0.00019106$
837	$+0.00484230$	$+0.00029143$	$+0.00026774$
841	$+0.00038073$	$+0.00004280$	$+0.00003954$
848	$+0.00010189$	-0.00005708	-0.00005644
856	-0.00040275	-0.00005387	-0.00005261
865	$+0.00216009$	-0.00006632	-0.00005151
875	$+0.00447180$	-0.00018830	-0.00019123
876	-0.00126093	$+0.00008691$	$+0.00008438$
880	$+0.00043030$	$+0.00003540$	$+0.00003336$
Mean	$+0.001242$	$+0.000031$	$+0.000029$

Absolute composed overlap remains unresolved. The two increments are smaller and have mixed signs. Within the declared linear pathway, these downstream diagnostics cannot repair the target-projection result or identify its cause. They do not constrain the independent direct-transfer contrast.

F.4 Technical-seed diagnostics

Optimization seeds test whether a fixed training procedure behaves reproducibly. They do not support biological generalization, because the same participants and stimuli are reused within each seed.

The brain-specific interpretation of the practical-utility evaluation depends on a reliable prerequisite contrast in controlled brain predictivity between the recorded-response arm and its matched permuted-response control. Table 23 reports the complete seed vector used to judge that prerequisite.

Table 23: Technical-seed values for the practical-utility prerequisite.

Seed	Target minus permuted-target representation contrast
0	$+0.063$
1	$+0.014$
2	$+0.004$
3	$+0.016$
4	$+0.005$
5	-0.006
6	-0.002
7	-0.005
Mean	$+0.011$
Median	$+0.0042$

The signs are split, and the mean is strongly influenced by seed 0. The seed rows are rounded independently to three decimal places; the displayed mean and median are computed from artifact-precision values. The complete vector documents mixed signs and seed-0 sensitivity; it does not establish a null or equivalence.

For the saved-student biological-transfer assay, each seed row averages participants and outer folds. The three columns use the same contrasts as Equation 12.

Table 24: Technical-seed values for saved-student biological transfer.

Seed	Δ^{T-X}	Δ^{T-K}	Δ^{X-K}
0	-0.00015727	-0.00035769	-0.00020042
1	$+0.00032614$	$+0.00003556$	-0.00029058
2	$+0.00004025$	$+0.00000892$	-0.00003133
3	$+0.00025080$	$+0.00015307$	-0.00009773
4	$+0.00018344$	-0.00006376	-0.00024721
5	$+0.00017470$	$+0.00014137$	-0.00003333
Mean	$+0.000136$	-0.000014	-0.000150

Before participant responses were inspected, the synthetic-target recovery, retained-student movement, and language-quality criteria were satisfied. The largest saved-arm language-quality difference, 0.002579, remained within the frozen 0.005 margin. The seed table therefore checks an interpretable manipulation while leaving biological inference to the participant table.

The exact-target seed diagnostics separate absolute target extraction from improvement over seed-matched ordinary KD and from composed-path increments. Within each seed, target extraction and incremental retention aggregate held-out target recovery; the two overlap increments average participants and outer folds.

Table 25: Technical-seed target-extraction and retention values.

Seed	Target extraction M_s	Incremental retention B_s
0	$+0.03965524$	$+0.00013878$
1	$+0.04096723$	$+0.00103098$
2	$+0.03932328$	-0.00159873
3	$+0.04291468$	-0.00234707
4	$+0.03920675$	-0.00038085
5	$+0.04228561$	-0.00197014
Mean	$+0.040725$	-0.000855

Table 26: Technical-seed composed-path increments.

Seed	Synthetic response minus KD D_s	Aligned minus twin J_s
0	-0.00003781	-0.00004017
1	-0.00004152	-0.00004668
2	-0.00001666	-0.00001488
3	$+0.00006481$	$+0.00006414$
4	$+0.00014778$	$+0.00014581$
5	$+0.00006938$	$+0.00006797$
Mean	$+0.000031$	$+0.000029$

Absolute target extraction is positive in every seed. Incremental retention has mixed signs and an interval containing zero. Both composed-path increments show the same uncertainty. The separation shows that fitting the target in absolute terms does not establish improvement over ordinary KD or transfer along the composed path.

Table 27: Primary retained-student movement values by technical seed.

Seed	Centered relative-Frobenius	Linear-CKA distance
0	0.186369	0.002633
1	0.216294	0.005457
2	0.298694	0.011778
3	0.195493	0.003585
4	0.197089	0.003925
5	0.196519	0.003423

The values use layer 6 and the same 1,999 untouched WikiText sentences in every arm; Equation 8 defines both distances.

No seed-only table is reported for the participant-specific recorded-response intervention. Its primary participant statistic is a median over crossed seed–block cells, so a seed-only vector cannot reconstruct the nonlinear participant aggregate in Equation 11. Seed variability remains inside each participant value, while the shared-block summary diagnoses the other crossed axis.

Table 28: Models in the cross-family language-quality analysis.

Model	Family	Parameters	Mid layer	Bits/byte	Unique R^2
Mistral-7B-v0.1	Mistral	7.24B	16	1.135	+0.0438
Qwen2.5-7B	Qwen	7.62B	14	1.152	+0.0235
Llama-3.2-3B	Llama	3.21B	14	1.163	+0.0464
Qwen2.5-3B	Qwen	3.09B	18	1.200	+0.0371
OPT-6.7B	OPT	6.66B	16	1.222	+0.0233
Pythia-2.8B	Pythia	2.78B	16	1.240	+0.0174
Qwen2.5-1.5B	Qwen	1.54B	14	1.242	+0.0297

Model	Family	Parameters	Mid layer	Bits/byte	Unique R^2
Llama-3.2-1B	Llama	1.24B	8	1.249	+0.0360
OPT-2.7B	OPT	2.65B	16	1.276	+0.0324
Pythia-1.4B	Pythia	1.41B	12	1.283	+0.0213
Pythia-1B	Pythia	1.01B	8	1.315	+0.0226
OPT-1.3B	OPT	1.32B	12	1.315	+0.0281
Qwen2.5-0.5B	Qwen	0.49B	12	1.342	+0.0356
GPT-2 Large	GPT-2	0.77B	18	1.351	+0.0173
Pythia-410M	Pythia	0.41B	12	1.381	+0.0233
GPT-2 Medium	GPT-2	0.35B	12	1.394	+0.0177
OPT-350M	OPT	0.33B	12	1.453	+0.0259
GPT-2 Small	GPT-2	0.12B	6	1.486	+0.0170
OPT-125M	OPT	0.13B	6	1.514	+0.0147
Pythia-160M	Pythia	0.16B	6	1.523	+0.0209
DistilGPT-2	GPT-2	0.08B	3	1.628	+0.0071
Pythia-70M	Pythia	0.07B	3	1.699	-0.0012

All 22 models enter the primary middle-layer correlation. The capable-model sensitivity retains the ten models below 1.30 bits per byte. The family-factor analysis excludes Llama and Mistral because each has fewer than three model sizes; those exclusions do not apply to the primary cross-family correlation.

These tables distinguish choices that changed or bounded an interpretation from checks of local stability. Participant heterogeneity and technical-seed variation answer different questions; neither can substitute for the other or expand the claims reported in Section **??**.

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